Extra Course D: Waveforms, Power, and Modulation

Fourier, RMS, PEP-to-average ratio, analog-to-digital conversion, modulation index, deviation ratio, OFDM, FDM, and TDM.

Groups E8A + E8B · 22 pool questions · 30-45 minute presentation · Foundation for Bites #7, #15 · Practice exams at eham.net/exams

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Part 1: Waveforms, Power, and ADC (E8A)

Fourier Analysis And Time vs Frequency

Fourier analysis shows that any periodic waveform can be decomposed into a sum of sine waves at the fundamental frequency and its harmonics. (E8A01) A **square wave** is made of a sine wave at the fundamental plus **only odd harmonics** (1f, 3f, 5f, 7f, …). The faster the rise time, the higher the harmonics extend.

There are two ways to look at any signal:

- **Time domain** shows **amplitude at different times** (E8A03) – the oscilloscope view.
- **Frequency domain** shows amplitude at each frequency – the spectrum analyzer view.

Fourier analysis is the math that converts between them.

RMS vs Average vs Peak

For a sine wave, all three values are related by constants: $V_{peak} = V_{RMS} \times \sqrt{2} \approx 1.414 \times V_{RMS}$, and $V_{RMS} \approx 0.707 \times V_{peak}$. But these constants ONLY hold for sine waves.

Benefit of a true-RMS calculating voltmeter: RMS is measured for both sinusoidal and non-sinusoidal signals (E8A05). An "averaging" meter just measures average and multiplies by 1.11 (a sine-wave constant). On distorted, modulated, or pulse waveforms that conversion is wrong – the true-RMS meter computes √(mean of v^2) directly and is right for any shape.

PEP vs Average Power For SSB

A voice SSB signal varies wildly with the speaker. There are loud peaks and quiet valleys. **Peak Envelope Power (PEP)** is the average power during one cycle at the loudest peak; **average power** is the long-term average over a speech sample.

Approximate PEP-to-average power ratio for an unprocessed SSB phone signal: 2.5 to 1 (E8A06).
What determines that ratio? Speech characteristics (E8A07). Different voices and microphone techniques produce different ratios – a compressed (processed) voice has a lower ratio.

Implication: a 100-watt-average SSB transmitter is pushing about 250 watts on voice peaks, which is what the 100-watt-class final amp actually has to handle without distortion.

Analog-to-Digital Conversion

An ADC samples a continuous voltage at fixed intervals and produces a binary number that represents the voltage at each sample.

Question	Answer	Q
Type of analog-to-digital conversion	**Successive approximation** (E8A02). The ADC tries each bit from MSB down, keeping it if the trial DAC output is still below the input.	E8A02
Why direct/flash ADCs in SDRs?	**Very high speed allows digitizing high frequencies** (E8A08). Flash converters use one comparator per output level so the conversion is instantaneous (but power-hungry and limited to 12 bits).	E8A08
How many input levels can an 8-bit ADC encode?	**256** (E8A09). $2^8 = 256$ distinct codes.	E8A09
Measure of ADC quality	**Total harmonic distortion** (E8A11). Captures both quantization and nonlinearity errors in one number.	E8A11
What is "dither"?	**A small amount of noise added to the input signal to reduce quantization noise** (E8A04). Counter-intuitive but real: dithering the input randomizes which way borderline samples round, turning periodic quantization patterns into harmless noise.	E8A04
Purpose of a low-pass filter at a DAC output	**Remove spurious sampling artifacts from the output signal** (E8A10). The DAC produces a staircase; the LPF (the reconstruction filter) smooths it back into a continuous signal.	E8A10

Part 2: Modulation, Multiplexing, and OFDM (E8B)

Modulation Index And Deviation Ratio (FM/PM)

These two numbers describe how "deep" frequency or phase modulation is. They look almost identical but have one key difference.

Modulation index (m) of an FM signal = **frequency deviation / modulating signal frequency** (E8B01). It changes with the audio frequency: low audio → big m, high audio → small m.
Deviation ratio (D) = **maximum frequency deviation / highest audio modulating frequency** (E8B09). It is the modulation index at the bandwidth-critical case (maximum deviation, highest audio); this defines how wide the signal gets via Carson's rule. Constant for a given system.
Phase modulation: the modulation index **does not depend on the RF carrier frequency** (E8B02). PM is set by phase shift in radians, independent of where the carrier sits.

E8B03: dev = 3000 Hz, audio = 1000 Hz → m = 3000/1000 = **3** (E8B03).
E8B04: dev = 6 kHz, audio = 2 kHz → m = 6/2 = **3** (E8B04).
E8B05: max swing = 5 kHz, highest audio = 3 kHz → D = 5/3 ≈ **1.67** (E8B05).
E8B06: max swing = 7.5 kHz, highest audio = 3.5 kHz → D = 7.5/3.5 ≈ **2.14** (E8B06).

Multiplexing: FDM, TDM, OFDM
Three ways to share a channel among multiple data streams.

Scheme	How it works	Q
Frequency Division Multiplexing (FDM)	**Dividing the transmitted signal into separate frequency bands that each carry a different data stream** (E8B10). Like several radio stations at different frequencies on the same antenna.	E8B10
Time Division Multiplexing (TDM)	**Two or more signals are arranged to share discrete time slots of a data transmission** (E8B11). One channel takes the first ms, another takes the next ms, etc.	E8B11
Orthogonal Frequency Division Multiplexing (OFDM)	**A digital modulation technique using subcarriers at frequencies chosen to avoid intersymbol interference** (E8B08). Many narrow subcarriers spaced just right so each carrier's nulls land on the other carriers' peaks – you can pack them tightly without crosstalk.	E8B08
What OFDM is used for in ham radio	**Digital modes** (E8B07). HF data modes and many VHF/UHF data systems use OFDM.	E8B07

Mental Cheat Sheet

- Fourier analysis: any periodic wave = sum of sine waves (fundamental + harmonics).
- Square wave = fundamental + **odd** harmonics only.
- Time domain = amplitude at different times.
- True-RMS meter reads correctly for non-sine waveforms.
- PEP/average ratio for unprocessed SSB ≈ **2.5**; set by **speech characteristics**.
- Successive approximation is a standard ADC type.
- Flash ADCs used in SDRs because of speed.
- 8-bit ADC → 2^8 = **256** levels.
- Total harmonic distortion measures ADC quality.
- Dither = small noise added to ADC input → reduces quantization noise.
- LPF after a DAC removes sampling-image artifacts (reconstruction filter).
- FM modulation index = deviation / modulating freq.
- Deviation ratio = max deviation / highest audio freq (constant for the system).
- PM modulation index does NOT depend on RF carrier frequency.
- FDM = different frequency bands.
- TDM = different time slots.
- OFDM = many subcarriers placed to be orthogonal (no intersymbol interference).

Common Traps

- Square wave is sine **plus odd harmonics only**. Not all harmonics, not even harmonics only.
- True-RMS meters matter for **non-sinusoidal** signals. The trap answer is "RMS noise factor."
- PEP-to-average is **speech characteristics**, not modulator type or amplifier gain.
- 8-bit ADC encodes **256** levels (2^8), not 8 levels.
- Dither **reduces** quantization noise; it is not an "abnormal condition."
- DAC reconstruction LPF removes **sampling artifacts**, not "out-of-sequence codes."
- Modulation index uses **modulating frequency**; deviation ratio uses **highest** audio frequency.
- Phase-modulation index is independent of carrier frequency.
- OFDM is **orthogonal subcarriers**, not "non-harmonically related FM" and not "narrow-band slow-speed."

Takeaway: Fourier links time and frequency domains; square waves are odd-harmonic sums. True-RMS meters handle distorted waveforms. SSB PEP/avg ratio is about 2.5 because of speech. ADC choices trade speed vs resolution; dither reduces quantization noise; LPF reconstructs the analog signal. FM modulation index changes with the modulating frequency, deviation ratio is its worst-case form. OFDM packs many orthogonal subcarriers; FDM splits by frequency; TDM splits by time.

Course D Question Bank (E8A + E8B)

E8A01. What technique shows that a square wave is made up of a sine wave and its odd harmonics?
A. Fourier analysis.
B. Vector analysis.
C. Numerical analysis.
D. Differential analysis.

E8A02. Which of the following is a type of analog-to-digital conversion?
A. Successive approximation.
B. Harmonic regeneration.
C. Level shifting.
D. Phase reversal.

E8A03. Which of the following describes a signal in the time domain?
A. Power at intervals of phase.
B. Amplitude at different times.
C. Frequency at different times.
D. Discrete impulses in time order.

E8A04. What is “dither” with respect to analog-to-digital converters?
A. An abnormal condition where the converter cannot settle on a value to represent the signal.
B. A small amount of noise added to the input signal to reduce quantization noise.
C. An error caused by irregular quantization step size.
D. A method of decimation by randomly skipping samples.

E8A05. What is the benefit of making voltage measurements with a true-RMS calculating meter?
A. An inverse Fourier transform can be used.
B. The signal’s RMS noise factor is also calculated.
C. The calculated RMS value can be converted directly into phasor form.
D. RMS is measured for both sinusoidal and non-sinusoidal signals.

E8A06. What is the approximate ratio of PEP-to-average power in an unprocessed single-sideband phone signal?
A. 2.5 to 1.
B. 25 to 1.
C. 1 to 1.
D. 13 to 1.

E8A07. What determines the PEP-to-average power ratio of an unprocessed single-sideband phone signal?
A. The frequency of the modulating signal.
B. Speech characteristics.
C. The degree of carrier suppression.
D. Amplifier gain.

E8A08. Why are direct or flash conversion analog-to-digital converters used for a software defined radio?
A. Very low power consumption decreases frequency drift.
B. Immunity to out-of-sequence coding reduces spurious responses.
C. Very high speed allows digitizing high frequencies.
D. All these choices are correct.

E8A09. How many different input levels can be encoded by an analog-to-digital converter with 8-bit resolution?
A. 8.
B. 8 multiplied by the gain of the input amplifier.
C. 256 divided by the gain of the input amplifier.
D. 256.

E8A10. What is the purpose of a low-pass filter used at the output of a digital-to-analog converter?
A. Lower the input bandwidth to increase the effective resolution.
B. Improve accuracy by removing out-of-sequence codes from the input.
C. Remove spurious sampling artifacts from the output signal.
D. All these choices are correct.

E8A11. Which of the following is a measure of the quality of an analog-to-digital converter?
A. Total harmonic distortion.
B. Peak envelope power.
C. Reciprocal mixing.
D. Power factor.

E8B01. What is the modulation index of an FM signal?
A. The ratio of frequency deviation to modulating signal frequency.
B. The ratio of modulating signal amplitude to frequency deviation.
C. The modulating signal frequency divided by the bandwidth of the transmitted signal.
D. The bandwidth of the transmitted signal divided by the modulating signal frequency.

E8B02. How does the modulation index of a phase-modulated emission vary with RF carrier frequency?
A. It increases as the RF carrier frequency increases.
B. It decreases as the RF carrier frequency increases.
C. It varies with the square root of the RF carrier frequency.
D. It does not depend on the RF carrier frequency.

E8B03. What is the modulation index of an FM phone signal having a maximum frequency deviation of 3000 Hz either side of the carrier frequency if the highest modulating frequency is 1000 Hz?
A. 3.
B. 0.3.
C. 6.
D. 0.6.

E8B04. What is the modulation index of an FM phone signal having a maximum carrier deviation of plus or minus 6 kHz if the highest modulating frequency is 2 kHz?
A. 0.3.
B. 3.
C. 0.6.
D. 6.

E8B05. What is the deviation ratio of an FM phone signal having a maximum frequency swing of plus or minus 5 kHz if the highest modulation frequency is 3 kHz?
A. 6.
B. 0.167.
C. 0.6.
D. 1.67.

E8B06. What is the deviation ratio of an FM phone signal having a maximum frequency swing of plus or minus 7.5 kHz if the highest modulation frequency is 3.5 kHz?
A. 2.14.
B. 0.214.
C. 0.47.
D. 47.

E8B07. Orthogonal frequency-division multiplexing (OFDM) is a technique used for which types of amateur communication?
A. Digital modes.
B. Extremely low-power contacts.
C. EME.
D. OFDM signals are not allowed on amateur bands.

E8B08. What describes orthogonal frequency-division multiplexing (OFDM)?
A. A frequency modulation technique that uses non-harmonically related frequencies.
B. A bandwidth compression technique using Fourier transforms.
C. A digital mode for narrow-band, slow-speed transmissions.
D. A digital modulation technique using subcarriers at frequencies chosen to avoid intersymbol interference.

E8B09. What is deviation ratio?
A. The ratio of the audio modulating frequency to the center carrier frequency.
B. The ratio of the maximum carrier frequency deviation to the highest audio modulating frequency.
C. The ratio of the carrier center frequency to the audio modulating frequency.
D. The ratio of the highest audio modulating frequency to the average audio modulating frequency.

E8B10. What is frequency division multiplexing (FDM)?
A. The transmitted signal jumps from band to band at a predetermined rate.
B. Dividing the transmitted signal into separate frequency bands that each carry a different data stream.
C. The transmitted signal is divided into packets of information.
D. Two or more information streams are merged into a digital combiner, which then pulse position modulates the transmitter.

E8B11. What is digital time division multiplexing?
A. Two or more data streams are assigned to discrete sub-carriers on an FM transmitter.
B. Two or more signals are arranged to share discrete time slots of a data transmission.
C. Two or more data streams share the same channel by transmitting time of transmission as the sub-carrier.
D. Two or more signals are quadrature modulated to increase bandwidth efficiency.

Answer Key

E8A01: A	E8A02: A	E8A03: B	E8A04: B	E8A05: D
E8A06: A	E8A07: B	E8A08: C	E8A09: D	E8A10: C
E8A11: A	E8B01: A	E8B02: D	E8B03: A	E8B04: B
E8B05: D	E8B06: A	E8B07: A	E8B08: D	E8B09: B
E8B10: B	E8B11: B			